



NATIONAL CHEMISTRY CONTEST - CUBA 2020

"It always seems impossible until it's done."

Nelson Mandela

Name (s): _____ Last name: _____
School: _____
Municipality: _____ Province: _____
CODE: _____ Exam: DAY 1. COMMON EXAM

Instructions

- Write all your data clearly in the indicated place (names, surnames, school, municipality, province). Do not type anything in the CODE field.
- Check that your agenda is complete. This is the agenda for the first day of the exam. It is made up of 5 questions (question 1, 15 items; question 2, 20 items; question 3, 6 items; question 4, 5 items; question 5, 11 items) and a total of 7 pages.

10th grade students only need to answer questions 1, 2, 3, and 4. 11th and 12th grade students only need to answer questions 1, 4, and 5.

- All necessary data, constants, conversions, and fundamental equations are included in the syllabus. You can use a scientific calculator.
- At the beginning of each question its value is indicated (out of a total of 100 points), as well as the value of each item (in marks).

Success!



Ministerio de Educación
República de Cuba



CONSTANTS, CONVERSIONS AND EQUATIONS

Avogadro's constant	$N_A = 6,0221 \cdot 10^{23} \text{ mol}^{-1}$	Ideal gas equation	$p \cdot V = n \cdot R \cdot T$
Gases contents	$R = 8,3145 \text{ J} \cdot \text{K}^{-1} \cdot \text{mol}^{-1}$	The third principle of thermodynamics	$Q = \Delta E + W$
Zeroth in the Celsius scale	273,15 K	Gibbs' energy	$\Delta G = \Delta H - T \cdot \Delta S$
Faraday's constant	$F = 96485 \text{ C} \cdot \text{mol}^{-1}$	Nernst's equation	$E = E^\circ - \frac{R \cdot T}{z \cdot F} \cdot \ln \frac{c_{\text{red}}}{c_{\text{ox}}}$
Ionic product of water at 25 °C	$K_W = 1,00 \cdot 10^{-14}$	Arrhenius' equations	$k = A \cdot \exp \left(-\frac{E_A}{R \cdot T} \right)$
Standard conditions	$p^\circ = 100 \text{ kPa}$ $c^\circ = 1 \text{ mol} \cdot \text{L}^{-1}$		$\Delta G^\circ = -R \cdot T \cdot \ln K^\circ = -z \cdot F \cdot \Delta E^\circ$

$$1 \text{ J} = 1 \text{ C} \cdot 1 \text{ V} = 1 \text{ kPa} \cdot 1 \text{ L} = 1 \text{ W} \cdot 1 \text{ s}$$

$$1 \text{ C} = 1 \text{ A} \cdot 1 \text{ s}$$

PERIODIC TABLE OF THE ELEMENTS

1 H 1.008	2 He 4.003											13 Al 26.98	14 Si 28.09	15 P 30.97	16 S 32.06	17 Cl 35.45	18 Ar 39.95
3 Li 6.94	4 Be 9.01											5 B 10.81	6 C 12.01	7 N 14.01	8 O 16.00	9 F 19.00	10 Ne 20.18
11 Na 22.99	12 Mg 24.31	3	4	5	6	7	8	9	10	11	12	13 Al 26.98	14 Si 28.09	15 P 30.97	16 S 32.06	17 Cl 35.45	18 Ar 39.95
19 K 39.10	20 Ca 40.08	21 Sc 44.96	22 Ti 47.87	23 V 50.94	24 Cr 52.00	25 Mn 54.94	26 Fe 55.85	27 Co 58.93	28 Ni 58.69	29 Cu 63.55	30 Zn 65.38	31 Ga 69.72	32 Ge 72.63	33 As 74.92	34 Se 78.97	35 Br 79.90	36 Kr 83.80
37 Rb 85.47	38 Sr 87.62	39 Y 88.91	40 Zr 91.22	41 Nb 92.91	42 Mo 95.95	43 Tc -	44 Ru 101.1	45 Rh 102.9	46 Pd 106.4	47 Ag 107.9	48 Cd 112.4	49 In 114.8	50 Sn 118.7	51 Sb 121.8	52 Te 127.6	53 I 126.9	54 Xe 131.3
55 Cs 132.9	56 Ba 137.3	57-71	72 Hf 178.5	73 Ta 180.9	74 W 183.8	75 Re 186.2	76 Os 190.2	77 Ir 192.2	78 Pt 195.1	79 Au 197.0	80 Hg 200.6	81 Tl 204.4	82 Pb 207.2	83 Bi 209.0	84 Po -	85 At -	86 Rn -
87 Fr -	88 Ra -	89-103	104 Rf -	105 Db -	106 Sg -	107 Bh -	108 Hs -	109 Mt -	110 Ds -	111 Rg -	112 Cn -	113 Nh -	114 Fl -	115 Mc -	116 Lv -	117 Ts -	118 Og -

57 La 138.9	58 Ce 140.1	59 Pr 140.9	60 Nd 144.2	61 Pm -	62 Sm 150.4	63 Eu 152.0	64 Gd 157.3	65 Tb 158.9	66 Dy 162.5	67 Ho 164.9	68 Er 167.3	69 Tm 168.9	70 Yb 173.0	71 Lu 175.0
89 Ac -	90 Th 232.0	91 Pa 231.0	92 U 238.0	93 Np -	94 Pu -	95 Am -	96 Cm -	97 Bk -	98 Cf -	99 Es -	100 Fm -	101 Md -	102 No -	103 Lr -

Problem 1. Miscellanea (15 puntos)

1.1	...	Total	Total
1	idem	15	15

Select the right answer in each case.

1.1. The air contains 21,0% in volume of O_2 . The percent of methane (CH_4) that would be necessary in the atmosphere to consume all the O_2 through complete combustion is:

A) 21,0%	B) 10,5%	C) 42,0%	D) cannot be calculated
----------	----------	----------	-------------------------

1.2. Calcium salts provide the flame with color:

A) orange	B) magenta	C) brick red	D) purple
-----------	------------	--------------	-----------

1.3. The electronic structure of manganese is:

A) $[Ar]3d^7$	B) $[Ar]3d^6 4s^1$	C) $[Ar]3d^5 4s^2$	D) $[Ar]3d^4 4s^2 4p^1$
---------------	--------------------	--------------------	-------------------------

1.4. It has the highest melting point:

A) Na	B) Li	C) K	D) Cs
-------	-------	------	-------

1.5. Substances with the same chemical bond:

A) N_2 , Na_2 , H_2	B) Au, CsAu, NaK	C) O_2 , SO_3 , ZnO_2	D) FeO, Al_2O_3 , SiO_2
---------------------------	------------------	-----------------------------	-----------------------------

1.6. It has the lowest boiling point:

A) I_2	B) Br_2	C) Cl_2	D) F_2
----------	-----------	-----------	----------

1.7. They are chemical species with the same valence shell electron count:

A) N_3^- , CON, COF	B) PH_3 , H_2O , HF	C) SF_4 , $SOCl_2$, IF_3	D) CO_2 , NO_2^- , NO_2^+
-----------------------	-------------------------	-------------------------------	---------------------------------

1.8. It has the highest secondary ionization energy:

A) Ca	B) He	C) O	D) Na
-------	-------	------	-------

1.9. It has a high ion polarization:

A) CsF	B) MgO	C) AuI_3	D) Na_2S
--------	--------	------------	------------

1.10. It is the least water-soluble substance:

A) CH_3CH_2OH	B) Na	C) HCl	D) SiO_2
-----------------	-------	--------	------------

1.11. The larger ion in the series P^{3-} , P^- , S^{2-} , S^- , Cl^- is:

A) S^-	B) P^-	C) Cl^-	D) P^{3-}
----------	----------	-----------	-------------

1.12. It has a very small electron affinity:

A) Be	B) B	C) K	D) Ga
-------	------	------	-------

1.13. It has the highest density:

A) Os	B) Cs	C) Hg	D) Pb
-------	-------	-------	-------

1.14. It possesses the smallest electric conductivity:

A) Pt	B) Si	C) H_2O	D) Hg
-------	-------	-----------	-------

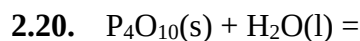
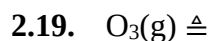
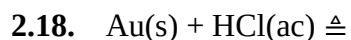
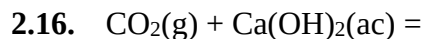
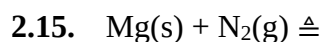
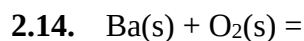
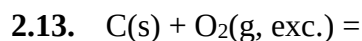
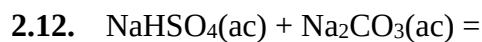
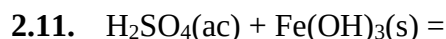
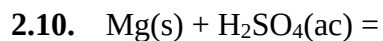
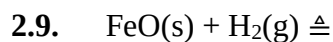
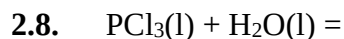
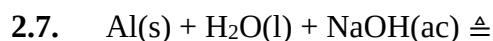
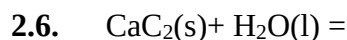
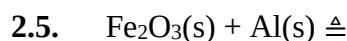
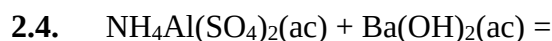
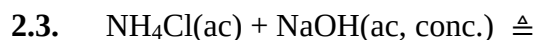
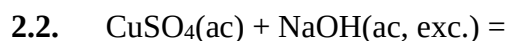
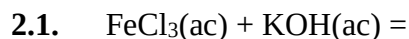
1.15. The oxidation number(s) of sulfur atoms in thiosulfate ion, $S_2O_3^{2-}$, is(are):

A) 1 y 3+	B) 2+	C) 3+	D) 0 y 4+
-----------	-------	-------	-----------

Problem 2. Chemical reactions (10 points)**[ONLY FOR 10th GRADE]**

2.1	...	Total	Total
1	idem	20	10

Complete and balance the following chemical equations. Take into account the aggregation states. Indicate when no reaction happens.

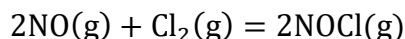


exc.: excess reagent; conc.: highly concentrated

Problem 3. Kinetics (10 points)**[ONLY FOR 10th GRADE]**

4.1	4.2	4.3	4.4	4.5	11.6	Total	Total
4	3	5	2	3	3	20	10

The reaction between nitric oxide and chlorine gives nitrosyl chloride as result. At 27 °C the following kinetic data was obtained.

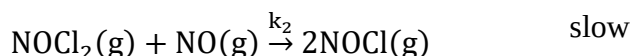
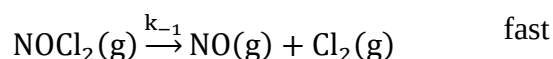


Experiment	Initial concentration ($\text{mol}\cdot\text{L}^{-1}$)		Initial rate ($\text{mol}\cdot\text{L}^{-1}\cdot\text{s}^{-1}$)
	NO	Cl_2	
1	0,020	0,010	$8,27\cdot 10^{-5}$
2	0,020	0,020	$1,65\cdot 10^{-4}$
3	0,020	0,040	$3,31\cdot 10^{-4}$
4	0,040	0,020	$6,60\cdot 10^{-4}$
5	0,010	0,020	$4,10\cdot 10^{-5}$

3.1. Find the rate law for the equation.

3.2. Calculate the average value of the rate constant.

It is possible that the reaction proceeds according to the mechanism



3.3. Use the steady-state approximation and find the rate law form the proposed mechanism.

3.4. Under what conditions the previous expression (item 3.3) matches the experimental rate law (item 3.1)?

3.5. It is known that the activation energy of the reaction is $22 \text{ kJ}\cdot\text{mol}^{-1}$. Calculate the rate constant when the temperature increases 10°C . If you cannot solve item 3.2 you may consider that $k_r = 50 \text{ u}$, where u symbolizes the measure units of the constant.

3.6. Represent the Lewis' formula of NOCl. Classify this molecule according to its polarity.

5.5. Chromium(III) chloride is obtained when metal chromium is treated with hydrochloric acid. Calculate the equilibrium constant for this reaction at 25 °C.

A galvanic cell was built aiming to study the dependency of the electrode potential of the redox couple $\text{Cr}^{3+}(\text{ac})/\text{Cr}(\text{s})$ with the pH value. A calomel electrode saturated with potassium chloride (ECS, $\text{Hg}_2\text{Cl}_2(\text{s})/\text{Hg}(\text{l})$) was used as reference electrode. The working electrode was prepared with 50,0 mL of a 0,0100 M CrCl_3 aqueous solution and a chromium wire as electronic conductor. A solution saturated in KNO_3 functioned as salt bridge. Next, slow addition of small volumes of 0,0500 M NaOH solution were added until a total of 40,0 mL was added to the original solution. Finally, a plot of galvanic cell voltage as a function of the volume of added sodium hydroxide was build.

5.6. Write the half equations of cathode and anode and the total cell equation.

5.7. Mention three characteristics that the electrolyte in the salt bridge must meet.

5.8. Estimate the initial emf of the cell. Neglect the hydrolysis of the Cr^{3+} ion.

5.9. Estimate the final emf of the cell. Neglect the possible formation of complex ions

Chromium deposition is usually performed by electrolysis of a solution of chromic acid. The anode is constituted by an alloy not reacting under the conditions employed, while the metallic object that should be coated works as cathode. An electrolytic cell was filled with 100 L of an aqueous solution containing 0,250 kg/L of CrO_3 . During the electrolysis, a current of 1500 A was applied for 10,0 h. The efficiency of the process was 15%.

5.10. Calculate the mass of chromium deposited in the cathode.

5.11. Write the half equations corresponding to two collateral redox processes that might happen in the previous electrolytic cell and provoke that the efficiency is lower than 100%.

Par redox	$\text{Hg}_2\text{Cl}_2(\text{s})/\text{Hg}(\text{l}), \text{KCl}(\text{sat})$	$\text{O}_2(\text{g})/\text{H}_2\text{O}(\text{l})$	$\text{H}^+(\text{ac})/\text{H}_2(\text{g})$
E°/V	0,271 V	1,23	0,000